

Architectural Particular Specification

FLASHING AND SHEET METAL

SECTION 6.3

Rev	Date	Description
0	14 APR 2022	First Issue

SECTION 6.3

FLASHING AND SHEET METAL

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Work of this Section includes, but not limited to, the furnishing and installing of the following:
1. Roof, roof curb, equipment plinth, parapet and coping, flashings, counter flashings, and trims.
 2. Shop fabricated scuppers, coping expansion joint covers, one piece splayed or right angle flashings, or counter flashings, and other specially fabricated sheet metal items as indicated in the Drawings.
 3. All sealants and ancillary material in joints between:
 4. Metal flashings, trims and accessories
 5. Flashing and Sheet Metal Work and adjoining materials or substrates.
 6. All required connection hardware and fasteners for attachment of the Work of this Section.
 7. Galvanic corrosion protection and isolation of dissimilar metals and incompatible substrates.
 8. Other sheet metal, flashings and trims as indicated in the Drawings and the Consulting Engineers' Contract Documents.
 9. Counter flashing at all plumbing, electrical, lighting protection, mechanical systems, equipments, vent stacks and other roof and wall penetrations.
 10. Temporary fire protection measure of flashing and sheet metal Work during installation of all torch-on applied modified bitumen roofing Work.
 11. Temporary occupational and construction safety measures provided, installed and maintained by the Main Contractor for the Project Work.
 12. Coordinate and interface with all Roofing Work in Section 6.1 to ensure weather tightness and warrantable work.
- B. Related Sections:
1. Section 3.1: UNIT MASONRY
 2. Section 4.1: STEEL AND METAL WORKS
 3. Section 6.1: DAMPPROOFING AND WATERPROOFING
 4. Section 6.5: JOINT SEALANTS
 5. Section 8.3: PAINTING
 6. Section 8.9: SPECIAL COATINGS
 7. Plumbing and Mechanical System and Equipments: Counter flashing between equipments, curbs or plinths and roof. Shop fabricated sheet metal flashing sleeves, collars for plumbing and mechanical items protruding through roof and wall assemblies. Refer to Consulting Engineers' Contract Documents.
 8. Electrical and Lightning Protection Systems and Equipments: Counter flashings between equipments and roof curbs or plinths. Shop fabricated sheet metal flashing sleeves and collars for electrical and lighting protection items protruding through wall and wall assemblies.

1.02 REFERENCES

A. British Standards

1. BS 219: 1977 (1989)-Specification for Soft Solders.
2. BS 729: 1986-Specification for Hot-Dip Galvanized Coatings on Iron and Steel Articles.
3. BS 1449: Part 2: 1983-Specification for Stainless and Heat-Resisting Steel Plate, Sheet and Strip.
4. BS 1470: 1972-Specification for Wrought Aluminium and Aluminium Alloys for General Engineering Purposes-Plate, Sheet and strip.
5. BS 3019: Part 1: 1984-Specification for TIG Welding of Aluminium, Magnesium and Their Alloys.
6. BS 3338: Part 12: 1961 (1990)-Methods for the Sampling of Solders.
7. BS 3571: Part 1: 1985-Specification for MIG Welding of Aluminium and Aluminium and Aluminium Alloys.
8. BS 4360: 1990-Specification for Weldable Structural Steels-Plates, Strip, Flats and Bars.
9. BS 6229: 1982-Code of Practice for Flat Roofs with Continuously Supported Coverings.
10. BS 6716 : 1986 (1992) : Guide to Properties and Types of Rubber
11. BS 7364: Part 1: 1991-Cathodic Protection. Code of Practice for Land and Marine Applications.
12. Code of Practice on Wind Effects-Hong Kong , 2004.
13. Occupational and Fire Safety Standards-BS EN 341, BS EN 344, BS EN 353, BS EN 361, BS EN 363, BS EN 367 and other applicable codes and standards.
14. PD 6484: 1979(1990)-Commentary on Corrosion at Bimetallic Contacts and Its Alleviation.

B. American Standards

1. American Architectural Manufacturers Association AAMA 605. 2-92: Voluntary Specification for High Performance Organic Coatings on Architectural Aluminium Extrusions and panels.
2. ASTM A167-93: Specification for Stainless and Heat Resisting Chromium-Nickel Steel Plate, Sheet and Strip
3. ASTM A525M-91b: Specification for sheet steel, Zinc Coated (Hot-Dip-Galvanized).
4. ASTM B209M-93: Specification for Aluminium and Aluminium Alloy Sheet Steel Plate.
5. ASTM B32-93: Specification for Solder Metal.
6. ASTM D3900 - 94 : Test methods for Rubber, Raw-Determination of Ethylene Units in EPM and EDPM.
7. U.S. Federal Specification FS-O-F506. Flux, Soldering, Paste and Liquid.
8. Sheet Metal and Air Conditioning Contractors' National Association : (SMACNA) Architectural Sheet Metal Manual, 5th Edition, 1993.
9. National Roofing Contractors' Association : NRCA Roofing & Waterproofing Manual, 4th Edition, 1996.

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

A. Performance Requirements

1. Wind Uplift:
 - a. All coping, fascia, metal edge, trim, counter flashing, cleat and fasteners exposed to weather and wind loading shall be installed and secured to resist wind uplift of minimum 1.5 Kpa. Design pressures, loads and up lift used for roof and sheet metal design shall be calculated in accordance with the requirement of CP3. The highest wind uplift value shall be used for the roof, sheetmetal, and flashing design and anchorage requirements.
 - b. Wind uplift pressure resistance at wind ward corners, roof edges, oversails, fascias, cappings, trims, drips and around all roof appurtenances shall be designed to minimum 2½ times the design wind uplift values per Clause 1.03 A (a) above.
2. Fastener Pullout Capacity:
 - a. All roofing and Sheet metal fasteners shall be corrosion-resistant, Type 316 Stainless Steel with serrated shank having a minimum pullout capacity of 91kg per fastener. No smooth shank common nails may be used. All corrosion-resistance fasteners shall meet U.S. Factory Mutual Standard 4470 requirements.
 - b. Pullout Tests: Due to the variety of physical conditions that can affect pullout resistance, on-site shall tests be conducted by an independent testing laboratory to determine actual pullout values.
 - c. The sections of the substrate where integrity is most in question shall be used for testing. Test areas shall include all corners perimeters and all areas of high wind pressure suction and up-lift.. The minimum number of pullout tests required is 12.
 - d. When new construction or other conditions prevent preliminary on-site pullout tests, the fastener manufacturer shall supply estimated pullout values for design and Tender purposes. On-site verification of the fastener pullout capacity shall be confirmed prior to system installation.
 - e. When tested pullout values do not meet the minimum fastener pullout capacity requirements, additional attachment shall be installed. Contact Fastener and Roof Manufacturer's Technical Services for additional information.
3. Details and installation of all flashing, sheet metal and proprietary aluminium composite panel coping and fixings shall accommodate all wind loads, wind uplift, thermal and building movement without material defects, determination, blow-out, dislodgment, pull-out or causing damage to waterproofing membranes, or sheetmetal. Expansion joints and weather tight joint covers shall be installed and detailed to meet all anticipated movements, wind pressures, loading and weather conditions.
4. Protect and isolate all dissimilar metals and non-compatible substrates, materials or finishes from all Work of this Section to prevent galvanic corrosion, staining and other damages.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

- A. The Flashing and Sheet metal and Fabricator (in submitting a Tender providing roofing and technical services, respectively, and executing and Project Work) will assume to warrant that all products, materials and items are suitable, and wind uplift resistance, thermal expansion and contraction, and weatherproof seal can be maintained in the various climatic and wind conditions encountered.

1.07 SUBMITTALS

- A. Shop Drawings.
 - 1. Indicate profiles, corner or splayed soldered or welded pieces, expansion joint provisions, sizes, connection attachments, reinforcing, anchorage, size and type of fasteners, and accessories.
 - 2. Include erection drawings, elevations and details where applicable. Layout and elevation drawings shall be drawn at 1:10 scale. Detail drawings at 1:2.5 scale.
 - 3. Indicate welded connections using standard welding symbols. Indicate net weld lengths.
 - 4. Prepare shop drawings under the supervision of a qualified registered Professional Engineer, if required by the Architect.
 - 5. Provide shop drawings for Exterior Wall Contractor, the Architect, the FSSC, the RSSC, Roofing Manufacturer, Main Contractor, and Consulting Engineer to review.
 - 6. Indicate soldered connections and fluxes used.
- B. Samples
 - 1. Samples of 200 mm x 200 mm size for each architectural metal, finish, coating and pattern specified herein. The cost of postage and transportation and all samples shall be the responsibility of the Contractor.
 - 2. Sample of each welding, butt joint, expansion joint cover, mitre joint and corner conditions of flashing and sheet metal work.
- C. Mill and test certificates of fasteners and architectural metals. Pullout capacity data on all fasteners to be used on the Project.
- D. Engineering calculations justifying flashing and sheet metal thicknesses, fixing and fastener design.
- E. Field testing reports.

1.08 DELIVERY, STORAGE, AND HANDLING

- A. Stack preformed material to prevent twisting, bending, or abrasion, and to provide ventilation. Cover all preformed material with removable protective film.
- B. Prevent contact with materials during storage which may cause discoloration, corrosion, staining, or damage.

- C. Store all flashing, sheet metal and preformed materials off ground and protected from the weather.

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

(NOT USED)

1.11 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. The organic coating formulator and qualified applicator shall jointly provide a fifteen (15) year warranty against material defect and workmanship for Work of this Section.
- C. Material and workmanship defects shall include, but not be limited, to chalking, corrosion, discoloration, blistering, fading, cracking, adhesion or cohesion failure, ultraviolet and ozone degradation during the warranty period. Dislodgement, warping, buckling, seam and joint failures, blow-out of any flashing, counter-flashing, counter-flashing and sheet metal pieces or cleats and fastener pull-outs.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PRODUCTS

1.13 MATERIALS

A. Flashing, Sheet Metal and Accessories

1. Stainless Steel Sheets, Strips, Bars and Plates: BS 1449: Part 2: 1983, and BS 4360. Type 316 with 2.5% Molybdenum minimum. Provide sheet of minimum 3.0 mm nominal thickness for exterior work.
2. Aluminium Alloy Plate, Sheet and Strips: BS 1470: 1983, acceptable 3003-H14 and 5005-H14 alloy and temper combinations for sheet and plate subject to fabrication, finish and structural requirements. Provide sheet of minimum 3.0 mm nominal thickness.
3. Stainless Steel Fasteners & Washers: Fasteners-BS 3111: Part 2: 1979, and BS 6105: 1981. Washers-BS 4320: 1968 and BS 4464: 1990.
4. Aluminium Fasteners: BS 1473: 1972.
5. Aluminium Extrusions : Aluminium Alloy AA 6063-T5, conforming to BS 1474 : 1987 : in mill finish, U.N.O.
6. Proprietary Penetration Flashing and Expansion Bellow Covers
7. Shims
 - a. At connections subject to thermal movement, or other movement, separate all pairs of moving surfaces with friction reducing pads. Pads shall have minimum 3 mm thickness, shall sufficiently reduce friction to permit movement, shall be resistant to wear, shall be positively retained in position (open ended slots are not acceptable). Pads shall not be subjected to heat damage from welding or cutting, or to excessive pressure from over-tightening of bolts.
 - b. Shims which transfer shear forces (tending to slide on shim against another) shall be steel plates, set in a staggered pattern and fillet welded to each other and to the adjacent steel surfaces. The shims and welds shall be structurally designed to support the applied loads.
 - c. Plastic shims are acceptable at static connections for which the shims transfer only compressive forces.
 - d. Wood shims are not acceptable.
8. Welding Materials: BS 639: 1986 and BS 5135: 1984, type required for materials being welded.
9. Hot-Dip Galvanised Sheet Sheets: BS2989:1992. Electroplated coating of zinc on steel sheet is not acceptable as substitute for hot-dip galvanised sheets.
10. Concrete Inserts: Cast iron or malleable bolts, washers, and shims.
11. Protective Coating Film: Strippable protective Polyvinyl Chloride(PVC) film of 0.3 mm thick shall be used to prevent damage to architectural metal surfaces during fabrication and erection.
12. Organic Coatings : PPG “ Duranar” factory - applied, containing, oven-baked polyvinylidene fluoride (PVF₂) finish, minimum 2 - coat system containing KYNAR 500 Resin meeting AAMA No. 605.2-92 requirements.
13. Protective Backing Paint: Bituminous, U.S. Federal Specification FSTT-C-494.
14. Waterproof Flashing Membrane :

Refer to Section 6.1
15. Solder: ASTM B32-93 use 50/50 for all applicable work, unless otherwise specified. Neutralize flux after soldering.

16. Soldering Flux: U.S. Federal Specification O-F-506, type best suited for specific metal.
17. Fasteners & Gasket Washers:
 - a. Stainless Steel Type 316 serrated shank fasteners meeting the requirements of U.S. Factory Mutual Standard 4470, BS 3111: Part 2, and BS 6105 with a minimum pullout resistance of 91 kg and a minimum effective embedment of 38 mm. No smooth shank nails may be used.
 - b. Watertight gasket with stainless steel washers conforming to BS 4320: 1968 and ASTM C864, solid dense gaskets shall have a Shore A hardness of 75+/-5 light color coated Santoprene minimum 1.6mm thick.
 - c. Fasteners, including concealed screws, nuts, bolts and other items required for connecting aluminium to aluminium or aluminium to steel shall be of non-magnetic stainless steel.
 - d. All fixing anchors, brackets, and similar attachments used in the erection shall be of aluminium, non-magnetic stainless steel, zinc coated steel or hot dip zinc galvanised steel.
18. Sealants and Ancillary Materials: Type as specified and provided in Section 4.1. The Contractor shall install sealants in all flashing and sheet metal lap joints and sealants in contact with flashing, coping counter-flashing, sheet metal and trims; and as indicated on the approved shop drawings.

B. Thicknesses of Flashing & Sheet Metal Items.

1. The thicknesses of all flashing, counter-flashing, sheet metal, and trims shall be determined by the most stringent of the following criteria :
 - a. Girth of sheet metal and flashing as measured by the total width of the sheet metal or flashing (when it is stretched out flat) and Table 2 at the end of this Section.
 - b. Required thickness to resist windload and wind uplift without deformation and other defects.

1.14 FABRICATION

A. Flashing and Sheet Metal Fabrication Requirements

1. Form sections true to shape, accurate in size, square, and free from distortion or defects.
2. Fabricate cleats, retainer clips, and starter strips of same material as sheet, interlockable with sheet. Cleats, retainer clips and starter strips to be minimum 1 gage thicker than sheet material.
3. Form pieces in longest practical lengths and not exceeding 3048 mm in length.
4. Hem exposed edges on underside minimum 13 mm, U.N.O. on the Drawings.
5. Form material with seams as indicated on approved shop drawings.
6. Solder and seal metal joints. After soldering, remove flux. Wipe and wash solder joint clean.
7. Fabricate outside and inside corners from single pieces with minimum 450mm long legs solder for rigidity.
8. Fabricate vertical faces with bottom edge formed outward at 30o , 6 mm in length and hemmed 25 mm to form drip.
9. Fabricate flashings to allow toe to extend 76 mm minimum over base flashing, or as dimensioned on the Drawings. Return and brake edges.

10. Pre-punch slotted holes on sheet metals and flashings at fastener locations on all flashing and sheet metal to allow thermal expansion and contraction and movements. Reprime all exposed cutted edges with zinc coating of minimum 610g/m³.

B. Welding and Soldering Requirements

1. Soldering

- a. Solder and seal flashing and sheet metal joints as required and indicated on the approved shop drawings. After soldering, remove flux. Wipe and wash solder joints clean.
- b. Do not solder aluminium.
- c. Perform soldering with well-heated coppers, so as to heat thoroughly the seam and sweat the solder through its full width. Roughen edges of stainless steel with emery cloth before soldering.
- d. Use type of solder flux compatible with the sheet metal. Remove acid flux residue by neutralizing with ammonia or baking soda and rinsing with clean potable water.
- e. Lap seams where soldered shall be finished not less than 25 mm wide.

2. Welding

- a. Welding of Aluminium:
 - 1) Welding of aluminium shall be done in the factory and in conformance to BS3019: Part 1 - 1984, BS3571: Part 1 - 1985, and BS3451:1981. Weld shall use inert gas process using an electric arc with a protective envelope of argon gas. Oxy-acetylene welding is not permitted as a substitute. Do not solder aluminium.
 - 2) Weld aluminium where specifically approved by the Architect.
 - 3) Welding of Alucobond Aluminium Composite Panels shall be shop performed by the Manufacturer.
- b. Welding of Steel : BS499: Part 1:1991 & Part 2:1989, and BS5135:1984.
- c. Welding of Stainless Steel : BS1453:1987 and BS2901: Part 2: 1990.
 - 1) All welding shall be done by electric fusion metal-arc method. Carbon-arc or gas welding will not be permitted. All welding shall be done in a thorough manner, with edging rod of same composition as sheets or parts welded. Welds shall be completed welds, strong and ductile, with excess metal ground off and joints finished smooth to match adjoining surface. Welds shall be free of mechanical imperfections such as gas holes, pits, runs, cracks, etc. and shall have same colour as adjoining surfaces.
- d. Grinding, Polishing and Finishing
 - 1) All welded exposed joints shall be suitably ground flush with adjoining material and neatly finished to harmonise therewith. Wherever material has been sunken or depressed by welding operation, such depressions shall be suitably hammered and peened flush to adjoining surface, and if necessary, again ground to eliminate low spots. All ground surfaces shall be smooth, consistent with good workmanship. Care shall be exercised in all grinding operations to avoid excessive heating of metal and metal discoloration. In all cases, grain of rough grinding shall be removed by successive polishing operations. Texture of final polishing operation shall be uniform and smooth, consistent with reasonable care and good workmanship.

- 2) Butt joints and contact joints, wherever they occur, shall be close fitting and shall not require solder as filler. Wherever brake bends occur, they shall be free of under extrudence and where such brakework does mar uniform appearance of material, all such marks shall be removed by suitable grinding, polishing and finishing. Wherever sheared edges occur, they shall be free of burrs, fins or irregular projections and shall be finished over such sheared edges. Where mitres or bull nosed corners occur, they shall be neatly finished with under edge of material neatly ground to uniform condition and in no case are overlapping materials to be acceptable.

1.15 FINISHES & FINISHING

A. Surface Preparation and Priming:

1. Clean surfaces of rust, scale, grease, and foreign matter prior to finishing.
2. Do not prime surfaces in direct contact bond with concrete or where field welding is required.
3. Prime paint items scheduled with one coat; touch up with same primer.
4. Hot-dip galvanise to BS3083: 1988 to minimum 600 g/m³ zinc coating (both sides) for all interior and exterior items specified or noted on the Drawings Alternatively, minimum ASTM A-527 G90 zinc coating thickness). Electroplated coating of zinc on steel or sheet metal is not permitted as a substitute.
5. Small areas of galvanised coating damaged by welding, cutting or by tough treatment during transit or erection shall be made good by they use of at least of two coats of zinc rich paint to BS4652:1979. Thoroughly clean all areas affected by welding with abrasive or mechanical driven tools before applying the zinc rich paint.
6. Backpaint all concealed metal surfaces with protective backing paint to a minimum dry film thickness of 15 mil.
7. Paint all exposed metal surfaces in accordance with Section 8.9 - Painting, and as indicated on Drawings.
8. Aluminium in contact with dissimilar metal (excluding non-magnetic stainless steel) shall be treated with an approved treatment for the prevention of electrolytic action and corrosion.
9. Aluminium surfaces in contact with mortar, concrete, plaster, masonry and similar materials and wet-applied materials in fire-proofing and absorptive materials shall be coated with an approved anti-corrosive, moisture-barrier material.

B. Stainless Steel Metal Finishes: BS1449: Part 2:1983, where scheduled herein, and on the Drawings.

1. Mill Finish
2. Hairline or Satin Finish
3. Mirror Polished finish

C. Organic Coating Finishes (PVF₂)

1. All Coatings, when cured, shall be visibly free of flow lines, streaks, sags, blisters or other surface imperfections.
2. All finishes shall match in gloss, and fall within the colour range of the approved samples.

3. Dry film thickness (DFT) of coating on exposed surface, when measured in accordance with BS 4842:1991, BS6496:1991, and BS6497:1991 shall not be less than 1.2 mil, except in channel recesses and internal corners, which shall be visually covered.
4. Prepare and submit colour samples to the Architect. After selection by the Architect, prepare 6 sets each of two samples which shall define the colour and gloss range and submit to the Architect for endorsement. Samples shall be identified and attached with a full laboratory report as per BS4842 BS6496, BS 6496, and BS6497.
5. Painted aluminium finish shall be factory oven cured, minimum two-coat finish U.N.O. of fluoropolymer resin supplied by a licensed formulator. The licensed formulator shall identify the finish which he supplies to meet the minimal requirements for formulation as set forth in the licensing agreement.
6. Application of the finish shall be carried out in accordance with the specifications issued by PPG Industries (licensed formulator) and a list of applicators specifically approved by PPG. The applicator shall provide written notification of approval by PPG prior to application of the finish. The formulation shall contain at least 70% fluoropolymer KYNAR 500 resin in the residual solids.
7. Pretreat metal surfaces in accordance with the procedure recommended by the manufacturer to provide proper surfaces for coating.
8. Field touch-up of painted aluminium is permitted only with written permission of the Architect. Unless touch-up is authorised, replace damaged material with new material.
 - a. Mill finish is not acceptable at structural silicone bonding surfaces.
 - b. Metal surface to which structural silicone will be adhered shall have a finish which demonstrates by test the ability to satisfy specified requirements.

PART 2 EXECUTION

2.01 EXAMINATION / INSPECTION

- A. Verify roof openings, cubs, pipes, sleeves, ducts, or vents through roof are solidly set, cant strips and reglets in place.
- B. Verify membrane termination and base flashings of roofing are in place, sealed, and secure.
- C. Coordinate installation of Coping, Fascia and Trims Work in Section 9.1. Refer and coordinate with Consulting Engineers' Contract Documents for other penetrations requiring flashing and sealing.
- D. Beginning of installation means acceptance of existing conditions and substrate.

2.02 PREPARATION

- A. Field measure site condition prior to fabrication work.
- B. Install starter and edge strips retainer clips and cleats before starting installation.

2.03 INSTALLATION

- A. All flashing and sheet metal work shall be installed in proper sequence in accordance to approved Shop Drawings to ensure weathertightness and warrantable Work.
- B. Install counterflashing true to lines and levels.
 - 1. Surface Mounted Flashings: Fill joint between flashing and wall or parapet with full bead of sealant and secure with continuous flat bar anchored to wall at maximum 300 mm on center, paralalled to roof line. Provide a continuous, full bead of sealant at top edge flashing. Sealant bead shall be of sufficient width to provide a 45 degree angle with vertical surface.
 - a. Provide minimum 76 mm end lap at continuous elevations.
 - b. Change in elevation of 100 mm provide 200 mm end lap.
 - c. Change in elevation of 200 mm provide 300 mm end lap.
- C. Counterflashing shall be installed in such a manner as to provide for continuous contact at base flashing with sufficient pressure at point of contact to prevent dislocation or blow-off due to wind uplift.
 - 1. Fabricate counterflashing as indicated on the approved shop drawings to provide weathertight seal within reglet installation.
 - 2. Lap joints minimum 76 mm, and seal with approved elastomeric sealant.
- D. Apply non-setting, compatible, waterproof plastic cement compound between metal flashings and modified bitumen flashings as recommended by Roofing Manufacturer.
- E. Fit flashings tight in place. Make corners square, surfaces true and straight in planes, and lines accurate to profiles.
- F. Solder metal joints watertight for full metal surface contact. After soldering, wash metal clean with neutralizing solution and rinse with water.

- G. Secure coping and exposed roof edge/fascia trim with continuous wind cleats staggerly fastened at 76 mm on center. Install retainer clips of 38 mm wide x 1.6 mm thick minimum at 458 mm on centers at roof side of coping and at counter flashings.
- H. Where indicated on the approved shop drawings, Flashing shall be let into walls a minimum of 25 mm, secured with lead wedges and pointed in waterproof cement mortar. Flashing shall be lapped and sealant sealed a minimum of 100 mm and the lower edge folded under for additional stiffness. Secure flashing with retainer clips.
- I. Parapet coping expansion joint and joint cover shall be installed maximum 3048 mm centers and fabricated per approved shop drawings. A 13 mm expansion gap with a 150 mm wide joint cover shall be installed at all expansion joint locations.
- J. Perform field welding in accordance with BS5135: 1984, and other standards referenced in Clause 2.02 C above. After installation, grind sight-exposed field welds smooth, touch-up welds, scratched or damaged surfaces with primer as noted in Clause 2.03 C above. The Employer's Project Manager reserve the right to test welded joints by ultrasonic method to BS 3923: Part 1: 1983 and Part 2: 1972. Should test results indicate that welds are below the standard of acceptance, carry out approved remedial measures and further acceptance tests. Remedial weld work and re-testing shall be at the expense of the Contractor.
- K. Avoid indirect and direct contact of aluminium with copper, brass, bronze or uncoated steel. Aluminium in contact with dissimilar metal (excluding non-magnetic stainless steel) shall be treated with an approved treatment for the prevention of electrolytic action and corrosion in accordance with BS 7364 and PD 6484.
- L. Aluminium surfaces in contact with mortar, concrete, plaster, masonry and similar materials and wet-applied materials in fire-proofing and absorptive materials shall be coated with an approved anti-corrosive, moisture-barrier material.
- M. Install proprietary Manville "Expand-O-Flash" penetration closures per Manufacturer's approved shop drawings.

2.04 CLEANING

- A. Clean surfaces with cleaning solution compatible with metal surfaces and approved by the FSSC and Alucobond Manufacturer.
- B. Use non-metallic tools in cleaning operations where recommended.

2.05 PROTECTION OF FINISHED WORK

- A. After flashing and sheet metal and exterior wall system are installed, it shall be the responsibility of the Main Contractor to verify that it is properly and adequately protected from damage. Suitable protection shall be provided wherever required which may not scratch stain or deface metal surfaces.
- B. All Flashing and Sheet Metal Work in progress shall be at all times during construction by use of a suitable strong, impervious protective film and board securely held in place and maintained. Additional protection shall be provided by the Main Contractor at traffical areas and during adverse weather conditions.

END OF SECTION